McMaster University ENGINEERING | mohawk | W BOOTH SCHOOL OF ENGINEERING PRACTICE AND TECHNOLOGY
McMaster-Mohawk Bachelor of Technology Partnership

Laboratory 3
Temperature Sensor

PROC TECH 3CC3

Cloud Computing and Internet of Things

B.Tech – PROCTECH

PRELAB

If you are planning to use your computer to perform the lab, make sure the following software is installed prior to this lab:

1. NI LabVIEW LINX Toolkit. Please refer to the following documents that are posted on A2L:
    a. **2 - LabVIEW Arduino Firmware Installation**
    b. **3 - LabVIEW Arduino Firmware Loading and Testing**

OBJECTIVES

In this lab you will learn the following:

1. Familiarize yourself the **LabVIEW environment.**
2. Build a **LabVIEW VI** that converts temperature in degrees Celsius to temperature in degrees Fahrenheit
3. Measure **thermistor sensor** readings with an Arduino and display the temperature with LabVIEW.

MATERIALS

1. Thermistor sensor
2. 10k Ohm Resistor
3. Breadboard
4. Arduino Mega
5. Jumper Wires

PROCEDURE

Exercise 0 – Testing Communication with Arduino

1. To make sure that all hardware and software components are working properly, please go to A2L>>Contents>>Labs>>Software Installation Instructions and implement the steps detailed in: **3 - LabVIEW Arduino Firmware Loading and Testing**

Exercise 1 – Convert °C to °F

Build a VI

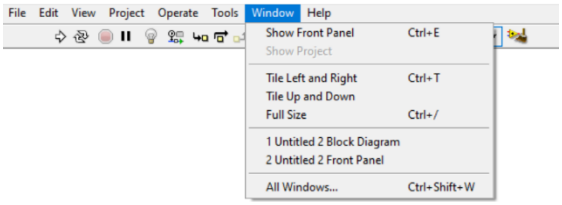
Complete the following steps to create a VI that takes a temperature in degrees Celsius and converts it to a temperature in degrees Fahrenheit.

Front Panel

1. Open LabVIEW and click **Create Project.**

2. Select a **Blank VI** and click finish.

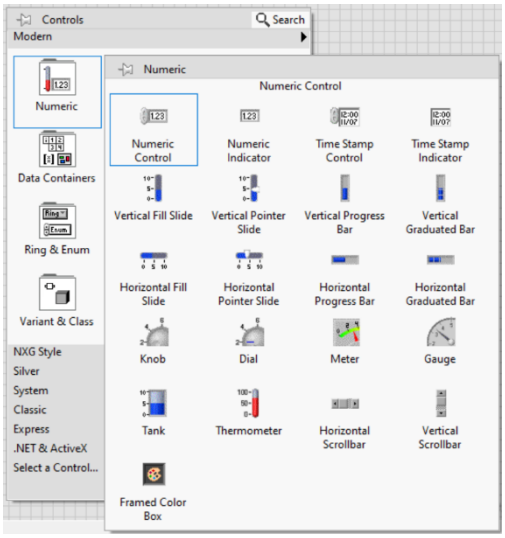
3. Two windows will open: `Front Panel` & `Block Diagram`. You can toggle between the two windows by clicking **Window»Show Block Diagram** or **Window»Show Fron Panel.**



4. (Optional) Select **Window»Tile Left and Right** to display the front panel and block diagram side by side.

5. Create a numeric digital control. You will use this control to enter the value for degrees Centigrade.

a. Open the **Controls** palette by *right-clicking* an open area on the front panel. Select the **Numeric Control** on the **Controls»Modern»Numeric** palette. If the **Controls** palette is not visible, you can open it by clicking **View»Controls Palette.**



b. Move the control to the front panel and click to place the control.
c. Type `deg C` inside the label and click outside the label or click the **Enter** button on the toolbar. If you do not type the name immediately, LabVIEW uses a default label. You can edit a label at any time by using the Labeling tool.

6. Create a numeric digital indicator. You will use this indicator to display the value for degrees Fahrenheit.
   a. Select **the Numeric Indicator** on the **Controls»Modern»Numeric** palette.
   b. Move the indicator to the front panel and click to place the indicator.
   c. Type `deg F` inside the label and click outside the label or click the Enter button.

7. LabVIEW creates corresponding control and indicator terminals on the block diagram. The terminals represent the data type of the control or indicator. For example, a DBL terminal represents a double-precision, floating-point numeric control, or indicator.

8. Note Control terminals have a thicker border than indicator terminals.

Block Diagram

1. Display the block diagram by clicking it or by selecting **Window»Show Diagram**.

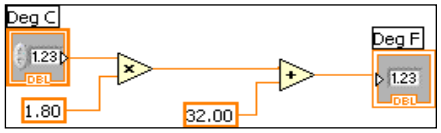


Figure 1: Deg C to Deg F block diagram

2. On the **block diagram**, right click to open the **Functions** palette. If the **Functions** palette is not visible, you can open it by clicking **View»Functions Palette.**

3. Select the **Multiply** and **Add** functions on the **Functions»Programming»Numeric** palette and place them on the block diagram.

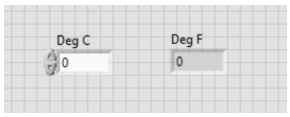
4. Select the **DBL numeric constant** on the **Functions»Programming»Numeric** palette and place two of them on the block diagram. When you first place the numeric constant, it is highlighted so you can type a value.

5. Type `1.8` in one constant and `32.0` in the other.

6. If you moved the constants before you typed a value, use the *Labeling tool* to enter the values.

7. Click on each function node(s) and drag the wires to connect the functions. Or use the *Wiring tool* in **View»Tools Palette»Connect Wire** to wire the icons as shown in **Figure 1.**
    a. You can bend a wire by clicking to tack the wire down and moving the cursor in a perpendicular direction. Press the spacebar to toggle the wire direction.
    b. When you move your mouse over a node, the node area blinks, indicating that clicking will connect the wire to that node and a tip strip appears, listing the name of the node.
    c. To cancel a wire you started, press the <Esc> key, right-click, or click the source node.

8. Display the front panel by clicking it or by selecting **Window»Show Front Panel**.



9. Save the VI because you will use this VI later in the course.
    a. Select **File»Save**.
    b. Navigate to `c:\exercises\LV Intro`.

    Note Save all the VIs you edit in this course in c:\exercises\LV Intro.

    c. Type `Convert C to F.vi` in the dialog box.
    d. Click the **Save** button.

10. Enter a number in the digital control and run the VI.
    a. Click on the digital control and type a new number.
    b. Click the **Run** button to run the VI.
    c. Try several different numbers and run the VI again.

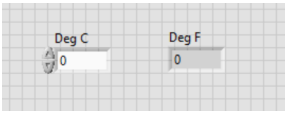
11. Select **File»Close** to close the `Convert C to F` VI.

End of Exercise

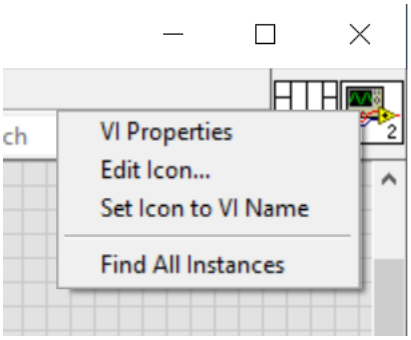
Exercise 2 – Create a SubVI

Front Panel

1. Select **File»Open** and navigate to the directory containing `Convert C to F` VI that was constructed in Exercise 1. The following front panel appears.



2. Right-click the icon in the upper right corner of the front panel and select **Edit Icon** from the shortcut menu. The **Icon Editor** dialog box appears.



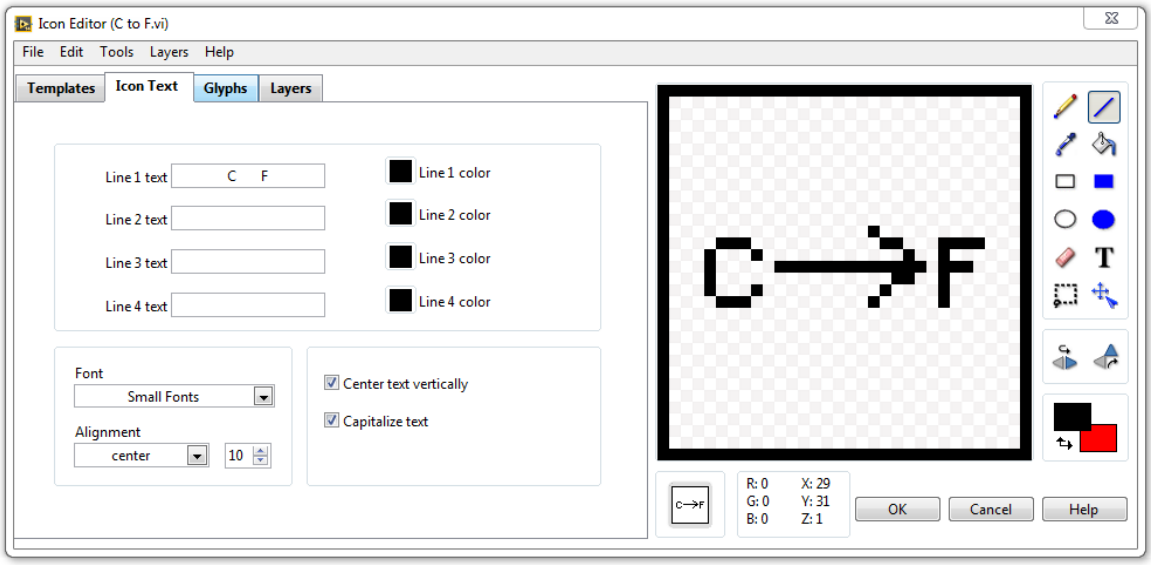
3. Double-click the Select tool  on the right side of the **Icon Editor** dialog box to select the default icon.



4. Click and drag to select the current icon. Press the <Delete> key to remove the default icon.

5. Double-click the Rectangle tool  to redraw the border.

6. Open the **Icon Text** tab to create the following icon.



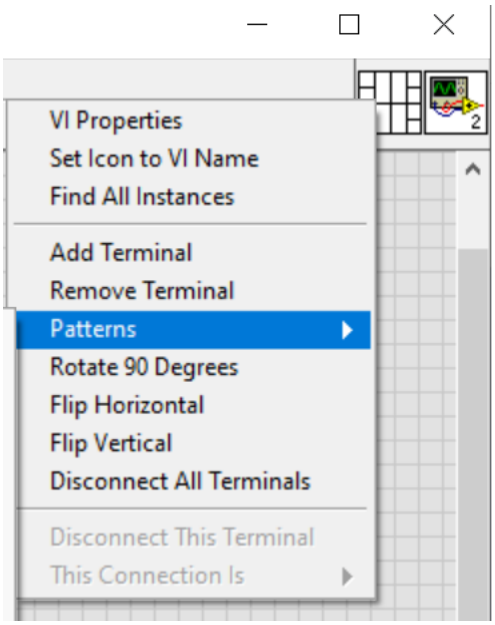
a. Type "C F" in Line 1 text. You might need to adjust the font size.

b. Use the line tool  to create the arrow.

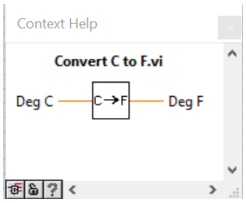
c. Use the Select tool and the arrow keys to move the arrow you created.

d. When the icon is complete, click the **OK** button to close the **Icon Editor** dialog box. The icon appears in the upper right corner of the front panel and block diagram.

7. Right-click the connectors area  icon in the upper right corner of the front panel. Select **Patterns** and then select a pattern with only two nodes  .



8. Assign the nodes to the digital control and digital indicator.
    a. Click the left node in the **connector pane**. The tool automatically changes to the Wiring tool, and the node turns **black**.
    b. Click the **deg C** control. The left node turns **orange**, and a marquee highlights the control.
    c. Click an open area of the front panel. The marquee disappears, and the node changes to the data type color of the control to indicate that you connected the node.
    d. Click the right node in the connector pane and click the **deg F** indicator. The right node turns orange.
    e. Click an open area on the front panel. Both nodes are orange  .
    f. Select **Help»Show Context Help** to display the **Context Help** window. Move the cursor over the connector pane. The **Context Help** window shows that both nodes are connected to floating-point values.



9. Select **File»Save** to save the VI because you will use this VI later in the course.

10. Select **File»Close** to close the Convert C to F VI.

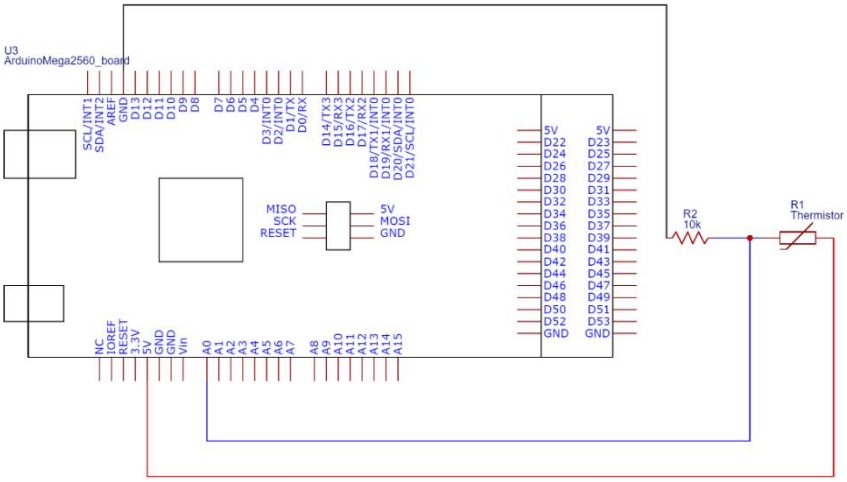
End of Exercise

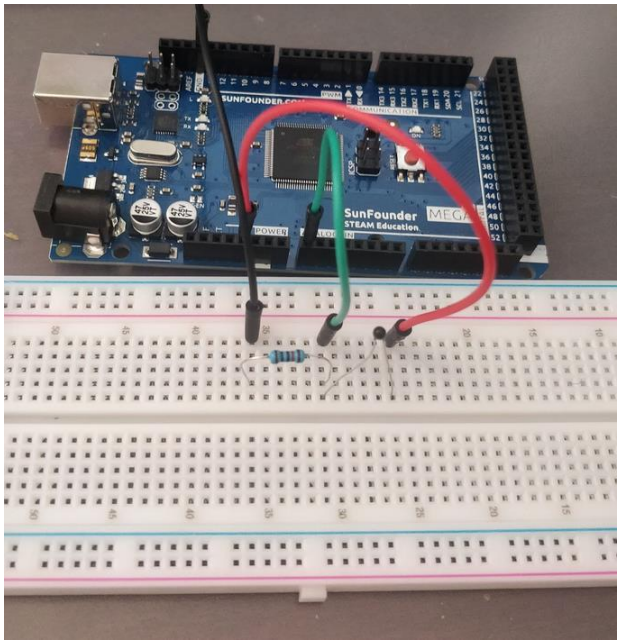
Exercise 3 - Data Acquisition

A thermistor is a temperature sensor. Its resistance drastically changes with the temperature. You will build a circuit, and to read the analog voltage values from the sensor to calculate the temperature.

Hardware

1. Connect the thermistor, 10k Ohm resistor, and Arduino as shown below. Use the 5V, GND and A0 pins of the Arduino. **Explain the purpose of this circuit in the report and discuss how temperature changes get converted to voltage changes that are read by the Arduino.**

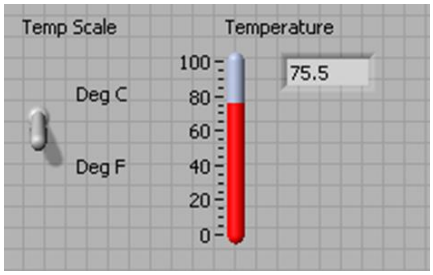




Front Panel

1. Select **File»New** to open a new front panel.

2. Create the thermometer indicator, as shown on the following front panel.



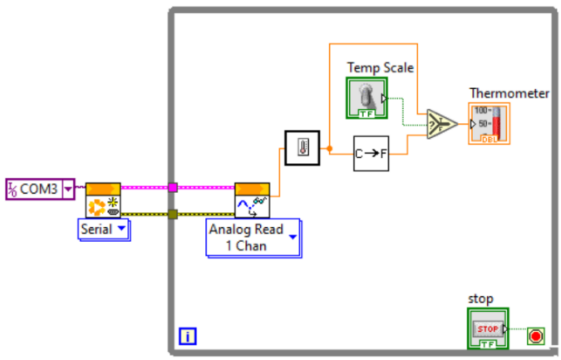
   a. Select the **Thermometer** on the **Controls»Modern»Numeric** palette and place it on the front panel.
   b. Type `Temperature` inside the label and click outside the label or click the **Enter** button on the toolbar.

c. Right-click the thermometer and select **Visible Items»Digital** Display from the shortcut menu to display the digital display for the thermometer.

3. Create the vertical switch control.
   a. Select the **Vertical toggle switch** on the **Controls»Modern»Boolean** palette.
   b. Type `Temp Scale` inside the label and click outside the label or click the **Enter** button.
   c. Use the *Labeling tool* or *double click* to place a free label, `deg C`, next to the TRUE position of the switch, as shown in the previous front panel.
   d. Place a free label, `deg F`, next to the FALSE position of the switch.

Block Diagram

1. Select **Window»Show Block Diagram** to display the block diagram.
2. Build the following block diagram. **In your report, explain the execution of the block diagram and the purpose of every block. What will happen if the while loop in the program is not used?**



   a. Place a **LINX Open** function from **Functions»MakerHub»LINX** palette.



   b. Place a **LINX Analog Read** function from **Functions»MakerHub» LINX»Peripherals »Analog** palette.
   c. Select **Functions»Select a VI…**, navigate to the **Convert C to F VI**, which you built in Exercise 2, and place the VI on the block diagram. This VI converts the Celsius readings to Fahrenheit.
   d. Select **Functions»Select a VI…**, navigate to the **lab_3_thermistor_sensor VI** that came with this lab**.** This VI converts the input analog *voltage* readings to *Celsius*. You may get a warning about the use of a **Mathscript node**. You can Ignore the warning.

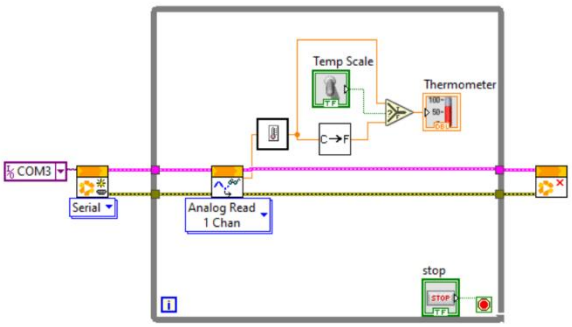
e. Place a **Select** function located on the **Functions»Programming»Comparison** palette. This function returns either the Fahrenheit (FALSE) or Celsius (TRUE) temperature value, depending on the value of **Temp Scale**.
f. Wire the output (right side) `Linx Resource` and `Error` nodes of the **LINX Open** to the input (left side) **LINX Analog Read** nodes.
g. Wire the rest of the functions as shown in the block diagram above. Note you must connect the `Voltage` output node of the **Analog Read** function to the `voltage` input node of the **lab_3_thermistor_sensor** function.
h. Right click on the `Serial Port` Input node of the **LINX Open** function. Select **Create Constant**. Open the **Constant** drop down menu and select the serial port your Arduino is connected to.



i. Right click on the `Analog Channel Input` node of the **LINX Analog Read** function. Select create constant. Set the constant to `0`.



3. Select a **While Loop** from the **Functions»Programming»Structures** palette. Place the loop to enclose the everything except for the **LINX Open** function.



a. Add a stop button to the `Loop Condition` node of the While Loop
b. Place a **LINX Close** function from **Functions»MakerHub»LINX** palette. Connect it to the **LINX Analog Read** as shown above.
c. (Optional) Click the **Clean Up Diagram**  on the top toolbar to tidy the diagram.

4. Display the front panel by clicking it or by selecting **Window»Show Panel**.

Make sure the Arduino is ON and connected over USB to your computer. Make sure the LINX firmware has been loaded to the Arduino!

5. Click the **Run** button to run the VI.

6. Put your finger on the temperature sensor and notice the temperature increase.

7. Click the **Stop** Button to stop the VI. **NOT THE ABORT BUTTON**

8. Select **File»Save** to save the VI. Choose a location on your hard drive and save the VI as Thermometer.vi.

9. Select **File»Close** to close the VI.

End of Exercise

EXTEND YOUR KNOWLEDGE

1. Add an additional thermometer to the front panel that will display the temperature in degrees Kelvin. You may create a new sub VI **OR** add functions directly to the block diagram.

SUBMISSION

1. Use your own words to describe the operation of the Thermistor sensor. Mention two applications of this sensor.
2. Answer all questions in this lab manual.
3. Include the new code corresponding to the extend your knowledge section and explain the block diagram modification that you implemented.
4. What are the benefits of using a subVI.
5. Submit your completed VI file with your student name and number on it.